



# UNIT 1

## CHAPTER 1: Operating Systems Overview

### 1.1 Definition

### 1.2 Evolution of Operating System

### 1.3 OS Components

- Process Management
- Main Memory Management
- File Management
- System Management
- Secondary Storage Management
- Protection System
- Command Interpreter System
- Information Management

### 1.4 Functions / Characteristics / Services of Operating System

- User's View and System's View
- Operating System Goals

### 1.5 Structure / Architecture of Operating System

- Simple Structure
- Layered Approach
- Microkernels
- Modules

### 1.6 Types of Operating Systems

- Batch Operating System
- Multiprogramming Operating System
- Multi-Tasking Operating System
- Time Sharing Operating System
- Real Time Operating System
- Distributed Operating System
- Network Operating System
- Multiprocessor (Parallel) Operating System
- Mobile Operating System

## **1.7 Operating System Services**

- User Interface
- Program Execution
- I/O Operations
- File System
- Communications
- Error Detection
- Resource Allocation
- Accounting
- Protection and Security

## **1.8 System Calls**

- Process Control
- File Management
- Device Management
- Information Maintenance
- Communication & Protection

## **1.9 System Programs**

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# **UNIT 2**

## **CHAPTER 2: Process Management**

### **2.1 Process Definition**

### **2.2 Process States**

- Primary Process States
- Additional Process States

### **2.3 Process State Transitions**

### **2.4 Process Scheduling**

- Scheduling Queues
- Schedulers
  - Long-Term Scheduler
  - Medium-Term Scheduler
  - Short-Term Scheduler
- Process Control Block (PCB)

## **2.5 Threads**

- Single Threaded and Multithreaded
- Multithreading Models
- Many-to-One
- One-to-One
- Many-to-Many
- Types of Threads

# **CHAPTER 3: CPU Scheduling**

## **3.1 Process Scheduling**

## **3.2 CPU Scheduler**

## **3.3 Scheduling Criteria**

## **3.4 Scheduling Objectives**

## **3.5 CPU Scheduling Algorithms**

- First Come First Serve (FCFS)
- Shortest Job First (SJF)
- Round Robin (RR)

## **3.6 Performance Evaluation**

- Deterministic Modeling
  - Queueing Models
  - Simulations
  - Implementation
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# **UNIT 3**

# **CHAPTER 4: Process Synchronization**

## **4.1 Process Synchronization**

## **4.2 Inter Process Communication (IPC)**

## **4.3 Mutual Exclusion**

## **4.4 Critical Section Problem**

- General Code
- Necessary Conditions

## **4.5 Solutions to Critical Section Problem**

- Two Process Solution

## **4.6 Race Conditions**

## **4.7 Semaphores**

- Mutual Exclusion with Semaphores
- Process Synchronization with Semaphores
- Counting Semaphores
- Implementing Semaphores

## **4.8 Monitors**

# **CHAPTER 5: Deadlocks**

## **5.1 Deadlock**

## **5.2 System Model**

## **5.3 Deadlock Characterization**

- Necessary Conditions
- Resource Allocation Graph

## **5.4 Methods for Handling Deadlock**

## **5.5 Deadlock Prevention**

## **5.6 Deadlock Avoidance**

- Safe State
- Avoidance Algorithms
- Resource Allocation Graph Method

## **5.7 Banker's Algorithm**

- Safety Algorithm
- Resource Request Algorithm
- Illustrative Example

## **5.8 Deadlock Detection**

- Single Instance
- Multiple Instances
- Detection Algorithm

## **5.9 Deadlock Recovery**

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# **UNIT 4**

## **CHAPTER 6: Memory & I/O Management**

### **6.1 Memory Management Background**

- Basic Memory Hardware
- Address Binding
- Logical vs Physical Address Space

### **6.2 Swapping**

### **6.3 Contiguous Memory Allocation**

- Fixed Partition (MFT)
- Variable Partition (MVT)

### **6.4 Paging**

### **6.5 Segmentation**

### **6.6 Virtual Memory**

### **6.7 Demand Paging**

- Hardware Support
- Advantages
- Performance

### **6.8 Page Replacement Algorithms**

- FIFO
- Optimal
- LRU
- LFU

## **6.9 Allocation of Frames**

- Minimum Frames
- Global vs Local Allocation

## **6.10 I/O Management**

## **6.11 Disk Management**

## **6.12 Polling**

## **6.13 Interrupts**

## **6.14 Direct Memory Access (DMA)**

## **6.15 Disk Structure**

## **6.16 Disk Scheduling Algorithms**